# Darwin Del Castillo, MD, MPH

darwin-del-castillo
ddelcastillo.f@gmail.com
github.com/ddelcastillof


## PROFESSIONAL SUMMARY

Physician and epidemiologist with an MPH in Epidemiology and applied health technology assessment experience, including cost-effectiveness modelling, budget impact analysis, and cost-of-illness estimation for a national payer. Skilled in evidence synthesis, from systematic reviews through meta-analysis, and in the medical statistics that supported HTA submissions, such as survival analysis, causal inference, and Bayesian modelling with formal uncertainty propagation. Experienced in generating real-world evidence from claims, electronic health records, and linked administrative data across the UK, US, and Peru. Seeking to apply this skill set to UK HTA submissions and market access within a global pharmaceutical environment.

## EDUCATION

**Master of Public Health (MPH) – Epidemiology** — **Sep 2023 – Jun 2025**
University of Washington — Seattle – USA

**Doctor of Medicine (MD)** — **Mar 2014 – Dec 2020**
Universidad Nacional Mayor de San Marcos — Lima – Peru

## ADDITIONAL EDUCATION

**Professional Certificate in Clinical Research** — **Sep 2021 – Apr 2022**
Universidad Peruana Cayetano Heredia — Lima – Peru

## SKILLS

**Programming:** R, Python, SQL (MySQL), SAS
**Cloud Computing:** Slurm HPC, Linux (bash)
**Tools:** Quarto, Shiny (R), REDCap, Git (GitHub, GitLab)
**Data Standards:** ICH E6, HIPAA, GDPR/UK GDPR, ICD-10/ICD-9, SNOMED CT
**Packages:** `tidyverse`, `survival`, `lme4`, `rstan`, `mice`, `cmprsk`, `ivreg`, `metafor`
**Statistical methods:** Bayesian hierarchical modelling (MCMC), GLM/GLMM, time-to-event models, marginal structural models, propensity score methods, competing risks, multiple imputation, meta-analysis, uncertainty estimation
**Health economics:** Cost-effectiveness and budget impact analysis, cost-of-illness estimation, decision-analytic modelling
**Languages:** English (fluent), Spanish (native)
**Core Competencies:** Real-world evidence generation, systematic review, observational study design, data management, stakeholder communication

## PROFESSIONAL EXPERIENCE

**Research Assistant – Epidemiology** — **Dec 2025 – Current**
School of Health and Wellbeing – University of Glasgow — Glasgow – UK

- Estimating causal effects of social policy on health outcomes using survey data with g-methods.
- Modelling the direct and indirect effects of economic determinants of health on all-cause mortality using linked administrative data and doubly robust inference.
- Conducting two systematic reviews for microsimulation parametrisation, from protocol and search strategy through screening and evidence synthesis.
- Simulating policy scenarios under uncertainty with a dynamic stochastic model, delivering dashboards to key stakeholders.

**Research Assistant – Clinical Informatics** **Jul 2024 – Jun 2025**
Institute for Health Metrics and Evaluation Seattle – USA

- Conducted a systematic review of descriptive epidemiological estimates to inform a Bayesian hierarchical model of burden of lower respiratory tract infections.
- Developed inputs for Bayesian models estimating the burden and clinical outcomes of lower respiratory tract infections worldwide, building MCMC pipelines that propagated uncertainty across 22 high-income countries.
- Analysed large-scale RWD, including structured EHR data from 2 countries with 60+ million records, identifying and maintaining the data sources underpinning epidemiological summaries of risk factors by sex.

**Health Technology Assessment Analyst** **Jan 2023 – Jan 2024**
Social Security System (ESSALUD) Lima – Peru

- Developed and adapted cost-effectiveness models comparing treatment strategies, informing national reimbursement decisions.
- Produced budget impact and cost-of-illness estimates from harmonised claims (>5 million encounters), quantifying the resource consequences of country-level coverage options.
- Generated real-world evidence inputs for economic models using ICD-10 phenotyping and probabilistic record linkage.

**Epidemiology Trainee** **Mar 2019 – Aug 2023**
CRONICAS Center of Excellence in Chronic Diseases Lima – Peru

- Conducted all stages of a systematic review – protocol, search strategy, screening, data extraction, and synthesis – informing an international core outcome set for multimorbidity trials (COSMOS, *BMJ Global Health*).
- Analysed longitudinal data from population cohorts and randomised trials in cardiometabolic epidemiology.
- Co-authored 2 scientific manuscripts involving longitudinal analyses and machine learning models.

## LICENSURE AND CERTIFICATION

| | |
|---|---|
| National Registry of Investigators, RENACYT-CONCYTEC, Investigator Level VII (P0160977) | **2024** |
| CITI Program, Good Clinical Practice (ICH E6), Credential ID 65079238 | **2023** |
| CITI Program, Human Subjects Learners, Credential ID 65079240 | **2023** |
| CITI Program, Biomedical Responsible Conduct of Research, Credential ID 65079239 | **2023** |
| Peruvian Medical Board, License No. 94534 | **2021** |

## PUBLICATIONS ORCID: 0000-0002-8609-0312

### Peer-Reviewed Publications

1. **Del Castillo-Fernández D**, Brañez-Condorena A, Villacorta-Landeo P, Saavedra-Garcia L, Bernabé-Ortiz A, Miranda J. Advances in the investigation of chronic non-communicable diseases in Peru. *An Fac med.* 2021; 81(4). DOI: 10.15381/anales.v81i4.18798
2. Lapidaire W, Proaño A, Blumenberg C, Loret de Mola C, Delgado CA, **Del Castillo D**, et al. Effect of preterm birth on growth and blood pressure in adulthood in the Pelotas 1993 cohort. *Int J Epidemiol.* 2023; 52(6):1870–7. DOI: 10.1093/ije/dyad084
3. Vidyasagaran AL, Ayesha R, Boehnke JR, Kirkham J, Rose L, Hurst JR, et al. Core outcome sets for trials of interventions to prevent and to treat multimorbidity in adults in low and middle-income countries: the COSMOS study. *BMJ Glob Health.* 2024; 9(8):e015120. DOI: 10.1136/bmjgh-2024-015120. Among authors: **Del Castillo Fernandez, D.**
4. Bazo-Alvarez JC, **Del Castillo D**, Piza L, Bernabé-Ortiz A, Carrillo-Larco RM, Smeeth L, et al. Multimorbidity patterns, sociodemographic characteristics, and mortality: Data science insights from low-resource settings. *Am J Epidemiol.* 2026; 194(4):901–11. DOI: 10.1093/aje/kwae466
5. Pinzon Cortes JA, Narongkiatikhun P, Fuhri Snethlage CM, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in T1D: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 1. *Endocrinol Metab Clin North Am.* 2026. DOI: 10.1016/j.ecl.2026.04.008
6. Pinzon Cortes JA, Narongkiatikhun P, Snethlage CF, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in T1D: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 2. *Endocrinol Metab Clin North Am.* 2026. DOI: 10.1016/j.ecl.2026.04.014